

Integrated Waveform and Mismatched Filters Design for Distributed DFRC System

Xue Yao

School of Electronics and Information
Engineering
Chongqing Sanxia University of Science
and Technology
Chongqing China
yaoxue@sanxiau.edu.cn

Junhao Zhang

School of Electronics and Information
Engineering
Chongqing Sanxia University of Science
and Technology
Chongqing China
1930621666@qq.com

Xianxiang Yu

School of Information and
Communication Engineering
University of Electronic Science and
Technology of China
Chengdu China
xianxiangyu@uestc.edu.cn

Abstract—This paper proposes a joint integrated waveform and mismatched filters design method for distributed Dual-Function Radar Communications (DFRC) system. Considering the scattering characteristics of extended targets in a distributed system, the integrated sidelobe level of the cross-correlation between multi-node echoes and mismatched filters is adopted as the optimization criterion. Envelope constraint is imposed due to hardware limitations, and an information modulation constraint is enforced to enable data transmission. To address the resulting non-convex and Non-deterministic Polynomial hard (NP-hard) optimization problem, an Iterative Block Coordinate Descent (IBCD) framework is adopted. This framework decomposes the original problem into several tractable subproblems, which are solved iteratively using the Coordinate Descent (CD) algorithm and Dinkelbach's algorithm (DA). Simulation results demonstrate that the proposed method can effectively reduce sidelobe levels while supporting multi-user communication.

Keywords—distributed Dual-Function Radar Communications (DFRC) system, joint integrated waveform and mismatched filters design, multi-user communication

I. INTRODUCTION

In recent years, the rapid development of wireless communication and radar sensing technologies has intensified the demand on limited spectrum resources. To address this challenge, Dual-Function Radar Communications (DFRC) system has received widespread attention [1]-[2]. In particular, waveform design is one of the key technologies enabling such systems [3]. Currently, integrated waveform design can be categorized into two main approaches: communication-centric dual-function waveform design [4]-[6] and radar-centric dual-function waveform design. This paper focuses on the latter one.

Some works consider the classical Differential Phase Shift Keying (DPSK) modulation to design integrated waveform. For instance, DPSK can be integrated with Linear Frequency Modulation (LFM) waveform to realize integrated DFRC waveform [7]-[8]. However, limited information modulation freedom leads to restricted communication data rates [1]. We have carried out extensive studies on integrated radar-communication waveform design using non-convex optimization theory. Representative methods, such as frequency-nulling modulation (FNM) [9] and ambiguity-function sidelobe-nulling modulation [10], have been developed for information embedding. However, to maintain detection performance, the number of nulling positions must be restricted, resulting in limited communication rates. To further improve the communication

rate, we have proposed a Position Index and Phase (PIP) modulation method to embed information [11]-[12].

However, most existing studies focus on phased-array or colocated MIMO systems and primarily consider narrowband signals for target detection, while relatively limited attention has been devoted to wideband integrated radar-communication waveform design in distributed systems. Distributed integrated systems can exploit target scattering characteristics from multiple spatial perspectives to achieve spatial diversity gains, which are beneficial for target detection. Besides, owing to high range resolution, strong anti-jamming capability, and low probability of intercept, wideband radar waveform design has been extensively investigated in the [13]-[14].

To address the research gap in distributed wideband DFRC waveform design, this paper proposes a joint design approach that considers the Degrees of Freedom (DoFs) at the receiver. Compared with wideband radar waveform design [13], the proposed method preserves the range resolution of wideband signals, reduces sidelobe levels, and enables multi-user communication.

II. OPTIMIZATION PROBLEM FORMULATION

Assume that the distributed DFRC system consists of N phased array nodes. Each node transmits a wideband signal to detect extended targets in the far field, while simultaneously transmitting communication information to the remaining $N-1$ nodes, as shown in Fig.1. Assume that the echo signal from the far-field extended target and the communication signals between nodes are separated in terms of reception time. In addition, the nodes achieve synchronization through mutual communication.

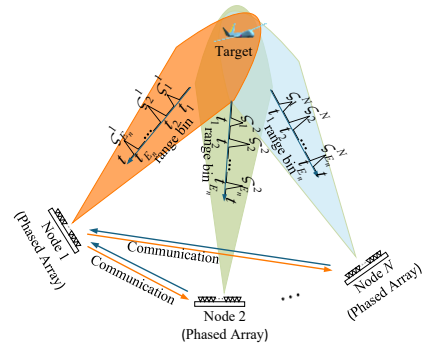


Fig.1. Detection scenario of a distributed DFRC system

Considering that the transmit waveforms at different nodes are orthogonal in the frequency domain, the multi-

node waveform transceiver design problem can be simplified to a joint design problem with a single transmitting waveform and multiple receiving-node filters.

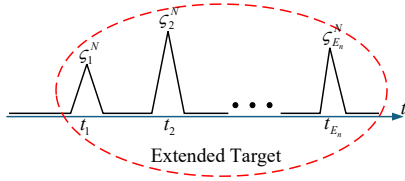


Fig.2. The scattering characteristics of extended targets

When detecting extended targets, the transmitted waveform $s(t)$ is scattered by the target and received by the N nodes. The extended target forms E_n scattering centers, with the scattering characteristics of each center represented by $\zeta_{e_n}^N, e_n = 1, 2, \dots, E_n$, as shown in Fig.2. The target impulse response $g_n(t)$ received by the n -th node can be expressed as

$$g_n(t) = \sum_{e=1}^{E_n} \zeta_{e_n}^N \delta(t - t_{e_n}) \quad (1)$$

where $\delta(t)$ represents the impulse response function, $t_{e_n} = t_0 + (e_n - 1)(c/2B)$, $t_0 = 2R_n/c$ and R_n is the radial distance of the extended target, B represents bandwidth, and c represents the speed of light.

Information is embedded in the transmitted waveform $s(t)$ through the PIP modulation technique [12]. The principle of information modulation is briefly recapped below. $s(t)$ is a pulse-based waveform with M subpulses. $N_p = N_{p_1} + N_{p_2} + \dots + N_{p_p}$ denote the number of communication subpulses, N_{p_i} represents the number of subpulses transmitted to node i , and N_R denote the number of subpulses used to optimization. a_{l_i} and a_{h_q} represent the amplitudes of the communication subpulses and the optimized subpulses, respectively, while ϕ_{l_i} and ϕ_{h_q} denote their corresponding phases. t_b denotes the subpulse duration. $s(t)$ can be modeled as

$$s(t) = \sum_{i=1}^{N_p} a_{l_i} e^{j\phi_{l_i}} \text{rect}\left[t - (l_i - 1)t_b / t_b\right] + \sum_{q=1}^{N_R} a_{h_q} e^{j\phi_{h_q}} \text{rect}\left[t - (h_q - 1)t_b / t_b\right] \quad (2)$$

where $\text{rect}(t/T) = 1$ when $0 \leq t \leq T$, otherwise $\text{rect}(t/T) = 0$. $\arg\{s(m)\}$ represents the phase of the m -th subpulse, $\mathcal{L}_n := \{\phi_{l_1}, \phi_{l_2}, \dots, \phi_{l_{N_p}}\}$ and $\bar{\mathcal{L}}_n := \{\phi_{h_1}, \phi_{h_2}, \dots, \phi_{h_{N_R}}\}$ correspond to the communication subpulse position set and non-communication subpulse position set, respectively. Ω_1 and Ω_2 represent the phase difference used for information transmission and optimization, respectively. Specifically, $\Omega_1 := \{\varphi_1, \varphi_2, \dots, \varphi_{N_p}\} := \{0, 2\pi/M_c, 2\pi/M_c, \dots, 2\pi(M_c - 1)/M_c\}$, $\varphi_i = \phi_i - \phi_{l_i}$, $\Omega_2 := \bigcup_{u=1}^{M_c} [l_u, h_u]$, $l_u = 2\pi(u - 1)/M_c + \kappa$, $h_u = 2\pi u/M_c - \kappa$, if the base of DPSK is W , then $M_c = PW$, P denotes the number of communication users, κ is the phase protection unit.

Taking the minimization of the ratio between the sum of sidelobe levels and the sum of mainlobe levels after mismatched filtering at all receiving nodes as the optimization criterion, and incorporating various practical constraints, a joint optimization problem for the wideband DFRC waveform and mismatched filters is formulated:

$$\mathcal{P}_1 \begin{cases} \min_{s, \{\mathbf{w}_n\}_{n \in N}} \sum_{n=1}^N \sum_{k=-L+1}^{L-1} |\nu_{n,k} \mathbf{r}_k(s, \mathbf{w}_n)|^2 / \sum_{n=1}^N |\mathbf{w}_n^H \mathbf{G}_n \mathbf{s}|^2 \\ \text{s.t.} \quad |s(m)| = 1, m = 0, 1, \dots, M-1 \\ \|\mathbf{w}_n\| = 1, n \in N \\ |\nu_{n,k} \mathbf{w}_n^H \mathbf{J}^k \mathbf{G}_n \mathbf{s}|^2 \leq \kappa_1, n \in N, -\bar{K} \leq k \leq \bar{K} \\ |\mathbf{w}_n^H \mathbf{G}_n \mathbf{s}|^2 \geq \kappa_2, n \in N \\ \arg\{s(1)\} = 0, \arg\{s(\mathcal{L}_n)\} \in \Omega_1, \arg\{s(\bar{\mathcal{L}}_n)\} \in \Omega_2 \end{cases} \quad (3)$$

where $\mathbf{w}_n$ denotes the discretized mismatched filter at the n -th node, $\mathbf{G}_n = \sum_{z=0}^{L-1} \mathbf{g}_n(z) \mathbf{J}_z$ denotes the pulse response matrix, $\mathbf{g}_n(z)$ represents the discretized representation of the target pulse response of the echo received by n -th node, and $\mathbf{J}_z$ denotes the shift matrix, $\mathbf{r}_k(s, \mathbf{w}_n) = \mathbf{w}_n^H \mathbf{J}^k \mathbf{G}_n \mathbf{s}$, where $\mathbf{r}$ represents the cross-correlation function, $\nu_{n,k}$ is the coefficient of the cross-correlation function $\mathbf{r}$, when $k = 0$, the coefficient $\nu_{n,k} = 0$, otherwise, $\nu_{n,k} = 1$. $\bar{K}$ represents the sidelobe level suppression range, κ_1 and κ_2 are positive real constants.

III. IBCD-DA ALGORITHM

This paper proposes an IBCD-DA algorithm, which alternately solves $\mathbf{s}$ and $\{\mathbf{w}_n\}_{n \in N}$. The subproblem related to $\mathcal{P}_{s^{(i)}}$ is solved using the CD algorithm, while the subproblem related to $\mathcal{P}_{\mathbf{w}_n^{(i)}}$ is solved using DA.

A. Given $\{\mathbf{w}_n^{(i-1)}\}_{n \in N}$, solve the following optimization problem $\mathcal{P}_{s^{(i)}}$

$$\mathcal{P}_{s^{(i)}} \begin{cases} \min_s \frac{\mathbf{s}^H \mathbf{C} \mathbf{s}}{\mathbf{s}^H \mathbf{D} \mathbf{s}} \\ \text{s.t.} \quad |s(m)| = 1, m = 0, 1, \dots, M-1 \\ \mathbf{s}^H \mathbf{C}_n^k \mathbf{s} - \kappa_1 \leq 0, n \in N, -\bar{K} \leq k \leq \bar{K} \\ \mathbf{s}^H \mathbf{D}_n \mathbf{s} - \kappa_2 \geq 0, n \in N \\ \arg\{s(1)\} = 0, \arg\{s(\mathcal{L}_n)\} \in \Omega_1, \arg\{s(\bar{\mathcal{L}}_n)\} \in \Omega_2 \end{cases} \quad (4)$$

$$\text{Let } \mathbf{C} = \sum_{n=1}^N \sum_{k=-L+1}^{L-1} \nu_{n,k} \mathbf{G}_n^H (\mathbf{J}^k)^H \mathbf{w}_n^{(i-1)} (\mathbf{w}_n^{(i-1)})^H \mathbf{J}^k \mathbf{G}_n, \mathbf{D} = \sum_{n=1}^N \mathbf{G}_n^H \mathbf{w}_n^{(i-1)} (\mathbf{w}_n^{(i-1)})^H \mathbf{G}_n, \mathbf{D}_n = \mathbf{G}_n^H \mathbf{w}_n^{(i-1)} (\mathbf{w}_n^{(i-1)})^H \mathbf{G}_n, \mathbf{C}_n^k = \nu_{n,k} \mathbf{G}_n^H (\mathbf{J}^k)^H \mathbf{w}_n^{(i-1)} (\mathbf{w}_n^{(i-1)})^H \mathbf{J}^k \mathbf{G}_n.$$

Applying the CD algorithm [15], $\mathcal{P}_{s^{(i)}}$ can be equivalently expressed as solving the following M optimization problems $\mathcal{P}_{s^{(i)}(m)}$:

$$\mathcal{P}_{s^{(i)}(m)} \begin{cases} \min_{s(m)} \frac{s(m)^H c_1^{(i-1)} s(m) + \Re\left\{\left(c_2^{(i-1)}\right)^H s(m)\right\} + c_3^{(i-1)}}{s(m)^H d_1^{(i-1)} s(m) + \Re\left\{\left(d_2^{(i-1)}\right)^H s(m)\right\} + d_3^{(i-1)}} \\ \text{s.t.} \quad |s(m)| = 1, \bar{n} \in N, m = 0, 1, \dots, M-1 \\ s(m)^H (c_{n,1}^k)^{(i-1)} s(m) + \Re\left\{\left((c_{n,2}^k)^{(i-1)}\right)^H s(m)\right\} + (c_{n,3}^k)^{(i-1)} \leq 0, n \in N, -\bar{K} \leq k \leq \bar{K} \\ s(m)^H (d_{n,1})^{(i-1)} s(m) + \Re\left\{\left((d_{n,2})^{(i-1)}\right)^H s(m)\right\} + (d_{n,3})^{(i-1)} \leq 0, n \in N \\ \arg\{s(1)\} = 0, \arg\{s(\mathcal{L}_n)\} \in \Omega_1, \arg\{s(\bar{\mathcal{L}}_n)\} \in \Omega_2 \end{cases} \quad (5)$$

where $c_1^{(t-1)} = \mathcal{X}_m^H \mathbf{C} \mathcal{X}_m$, $c_2^{(t-1)} = 2\mathcal{X}_m^H \mathbf{C}^H \bar{\mathbf{s}}_m^{(t-1)}$, $c_3^{(t-1)} = (\bar{\mathbf{s}}_m^{(t-1)})^H \mathbf{C}^H \bar{\mathbf{s}}_m^{(t-1)}$, $d_1^{(t-1)} = \mathcal{X}_m^H \mathbf{D} \mathcal{X}_m$, $d_2^{(t-1)} = 2\mathcal{X}_m^H \mathbf{D}^H \bar{\mathbf{s}}_m^{(t-1)}$, $d_3^{(t-1)} = (\bar{\mathbf{s}}_m^{(t-1)})^H \mathbf{D}^H \bar{\mathbf{s}}_m^{(t-1)}$, $\bar{\mathbf{s}}_m^{(t-1)} = \mathbf{s}^{(m)} - \mathcal{X}_m \mathbf{s}(m)$, $(c_{n,1}^k)^{(t-1)} = \mathcal{X}_m^H \mathbf{C}_n^k \mathcal{X}_m$, $(c_{n,2}^k)^{(t-1)} = 2\mathcal{X}_m^H (\mathbf{C}_n^k)^H \bar{\mathbf{s}}_m^{(t-1)}$, $(c_{n,3}^k)^{(t-1)} = (\bar{\mathbf{s}}_m^{(t-1)})^H (\mathbf{C}_n^k)^H \bar{\mathbf{s}}_m^{(t-1)} - \kappa_1$, $(d_{n,1})^{(t-1)} = -\mathcal{X}_m^H \mathbf{D}_n \mathcal{X}_m$, $(d_{n,2})^{(t-1)} = -2\mathcal{X}_m^H (\mathbf{D}_n)^H \bar{\mathbf{s}}_m^{(t-1)}$, $(d_{n,3})^{(t-1)} = -(\bar{\mathbf{s}}_m^{(t-1)})^H (\mathbf{D}_n)^H \bar{\mathbf{s}}_m^{(t-1)} + \kappa_2$, $\mathcal{X}_m \in \mathbb{C}^M$, if $\bar{i} = m$, $\mathcal{X}_m(\bar{i}) = 1$, otherwise $\mathcal{X}_m(\bar{i}) = 0$, $\mathbf{s}^{(m)} = [\mathbf{s}^{(t)}(1), \dots, \mathbf{s}^{(t)}(m), \mathbf{s}^{(t-1)}(m+1), \dots, \mathbf{s}^{(t-1)}(M)]^T \in \mathbb{C}^M$.

Let $\varpi = \tan(\phi_m/2)$ and denote ϕ_m as the phase of $\mathbf{s}(m)$, then the objective function of optimization problem $\mathcal{P}_{s^{(t)}(m)}$ can be expressed as:

$$\mathcal{P}_{s^{(t)}(m)} \begin{cases} \min_{\varpi} & \frac{\tilde{c}_1^{(t-1)} \varpi^2 + \tilde{c}_2^{(t-1)} \varpi + \tilde{c}_3^{(t-1)}}{\tilde{d}_1^{(t-1)} \varpi^2 + \tilde{d}_2^{(t-1)} \varpi + \tilde{d}_3^{(t-1)}} \\ \text{s.t.} & (\tilde{c}_{n,1}^k)^{(t-1)} \varpi^2 + (\tilde{c}_{n,2}^k)^{(t-1)} \varpi + (\tilde{c}_{n,3}^k)^{(t-1)} \leq 0, \\ & n \in N, -\bar{K} \leq k \leq \bar{K} \\ & (\tilde{d}_{n,1})^{(t-1)} \varpi^2 + (\tilde{d}_{n,2})^{(t-1)} \varpi + (\tilde{d}_{n,3})^{(t-1)} \leq 0, \\ & n \in N \\ & \phi_m \in \Phi_m \end{cases} \quad (6)$$

where $\tilde{c}_1^{(t-1)} = c_3^{(t-1)} + c_1^{(t-1)} - \Re\{c_2^{(t-1)}\}$, $\tilde{c}_2^{(t-1)} = 2\Im\{c_2^{(t-1)}\}$, $\tilde{c}_3^{(t-1)} = c_3^{(t-1)} + c_1^{(t-1)} + \Re\{c_2^{(t-1)}\}$, $\tilde{d}_1^{(t-1)} = d_3^{(t-1)} + d_1^{(t-1)} - \Re\{d_2^{(t-1)}\}$, $\tilde{d}_2^{(t-1)} = 2\Im\{d_2^{(t-1)}\}$, $\tilde{d}_3^{(t-1)} = d_3^{(t-1)} + d_1^{(t-1)} + \Re\{d_2^{(t-1)}\}$, $(\tilde{c}_{n,1}^k)^{(t-1)} = (c_{n,3}^k)^{(t-1)} + (c_{n,1}^k)^{(t-1)} - \Re\{(c_{n,2}^k)^{(t-1)}\}$, $(\tilde{c}_{n,2}^k)^{(t-1)} = 2\Im\{(c_{n,2}^k)^{(t-1)}\}$, $(\tilde{c}_{n,3}^k)^{(t-1)} = (c_{n,3}^k)^{(t-1)} + (c_{n,1}^k)^{(t-1)} + \Re\{(c_{n,2}^k)^{(t-1)}\}$, $(\tilde{d}_{n,1})^{(t-1)} = (d_{n,3})^{(t-1)} + (d_{n,1})^{(t-1)} - \Re\{(d_{n,2})^{(t-1)}\}$, $(\tilde{d}_{n,2})^{(t-1)} = 2\Im\{(d_{n,2})^{(t-1)}\}$, $(\tilde{d}_{n,3})^{(t-1)} = (d_{n,3})^{(t-1)} + (d_{n,1})^{(t-1)} + \Re\{(d_{n,2})^{(t-1)}\}$. Φ_m is the set of phases that satisfy the communication constraints. The $\mathcal{P}_{s^{(t)}(m)}$ can be solved using the method outlined in the [17].

B. Given $\mathbf{s}^{(t)}$, sequentially solve optimization problems

$\mathcal{P}_{\mathbf{w}_n^{(t)}}$.

$$\mathcal{P}_{\mathbf{w}_n^{(t)}} \begin{cases} \min_{\mathbf{w}_n} & \frac{\mathbf{w}_n^H \mathbf{E} \mathbf{w}_n + e^{(t-1)}}{\mathbf{w}_n^H \mathbf{F} \mathbf{w}_n + f^{(t-1)}} \\ \text{s.t.} & \|\mathbf{w}_n\| = 1 \\ & \mathbf{w}_n^H \mathbf{E}^k \mathbf{w}_n - \kappa_1 \leq 0, -\bar{K} \leq k \leq \bar{K} \\ & \mathbf{w}_n^H \mathbf{F} \mathbf{w}_n - \kappa_2 \geq 0 \end{cases} \quad (7)$$

where $\mathbf{E} = \sum_{k=-L+1}^{L-1} \mathbf{v}_{n,k} \mathbf{J}^k \mathbf{G}_n \mathbf{s}^{(t)} (\mathbf{s}^{(t)})^H \mathbf{G}_n^H (\mathbf{J}^k)^H$, $\mathbf{F} = \mathbf{G}_n \mathbf{s}^{(t)} (\mathbf{s}^{(t)})^H \mathbf{G}_n^H$, $e^{(t-1)} = \sum_{n \neq n} \sum_{k=-L+1}^{L-1} \mathbf{v}_{\bar{n},k} (\mathbf{w}_{\bar{n}}^{(t-1)})^H \mathbf{J}^k \mathbf{G}_{\bar{n}} \mathbf{s}^{(t)} (\mathbf{s}^{(t)})^H \mathbf{G}_{\bar{n}}^H \mathbf{w}_{\bar{n}}^{(t-1)}$, $f^{(t-1)} = \sum_{n \neq n} (\mathbf{w}_{\bar{n}}^{(t-1)})^H \mathbf{G}_{\bar{n}} \mathbf{s}^{(t)} (\mathbf{s}^{(t)})^H \mathbf{G}_{\bar{n}}^H \mathbf{w}_{\bar{n}}^{(t-1)}$, $\mathbf{E}^k = \mathbf{v}_{n,k} \mathbf{J}^k \mathbf{G}_n \mathbf{s}^{(t)} (\mathbf{s}^{(t)})^H \mathbf{G}_n^H (\mathbf{J}^k)^H$.

Based on generalized fractional programming theory, auxiliary variables are introduced to transform the objective function into the following form:

$$\zeta(y, \mathbf{w}_n) = g_0(\mathbf{w}_n) - y g_1(\mathbf{w}_n) \quad (8)$$

where $g_1(\mathbf{w}_n) = \mathbf{w}_n^H \mathbf{E} \mathbf{w}_n + e^{(t-1)}$, $g_0(\mathbf{w}_n) = \mathbf{w}_n^H \mathbf{F} \mathbf{w}_n + f^{(t-1)}$. According to the DA [16], the solution process can be summarized as follows:

Given $\mathbf{w}_n^{(t-1)}$, it follows that $y^{(t)} = g_0(\mathbf{w}_n^{(t-1)}) / g_1(\mathbf{w}_n^{(t-1)})$.

Given $y^{(t)}$, $\mathbf{w}_n$ is updated by solving the following approximate optimization:

$$\mathcal{P}'_{\mathbf{w}_n} \begin{cases} \max_{\mathbf{w}_n} & \Re\left\{\left(2\mathbf{F}_n^H \mathbf{w}_n^{(t-1)}\right)^H \mathbf{w}_n\right\} - \left(\mathbf{w}_n^{(t-1)}\right)^H \mathbf{F}_n \mathbf{w}_n^{(t-1)} - y^{(t)} \mathbf{w}_n^H \mathbf{E}_n \mathbf{w}_n + \left(f_n^{(t-1)} - y^{(t)} e_n^{(t-1)}\right) \\ \text{s.t.} & \mathbf{w}_n^H \mathbf{w}_n \leq 1 \\ & \Re\left\{\left(-2\mathbf{w}_n^{(t-1)}\right)^H \mathbf{w}_n\right\} + \left(\mathbf{w}_n^{(t-1)}\right)^H \mathbf{w}_n^{(t-1)} + 1 \leq 0 \\ & \mathbf{w}_n^H \mathbf{E}^k \mathbf{w}_n - \kappa_1 \leq 0, -\bar{K} \leq k \leq \bar{K} \\ & \kappa_2 - \Re\left\{\left(2\mathbf{F}^H \mathbf{w}_n^{(t-1)}\right)^H \mathbf{w}_n\right\} + \left(\mathbf{w}_n^{(t-1)}\right)^H \mathbf{F} \mathbf{w}_n^{(t-1)} \leq 0 \end{cases} \quad (9)$$

Solve using the CVX toolbox and repeat the above process until convergence is achieved.

TABLE I. ALGORITHM FLOW

Algorithm1: Procedure of IBCD-DA algorithm	
Input:	$\mathbf{s}, \mathbf{w}, \zeta, \mathbf{G}, N, M, \kappa_1, \kappa_2, \bar{K}, \Omega_1, \Omega_2, \varepsilon$;
Output:	$\mathbf{s}^*, \mathbf{w}^*$;
1:	$t = 0$, $\mathbf{s}^{(0)}$, $\mathbf{w}_n^{(0)}$;
2:	$t = t + 1$;
3:	Fixing $\mathbf{w}_n^{(t-1)}$, $\mathcal{P}_{s^{(t)}}$ is solved using the CD algorithm;
4:	Fixing $s^{(t)}$, $\mathcal{P}_{\mathbf{w}_n^{(t)}}$ is solved based on the DA;
5:	Calculate the current objective function value $Obj^{(t)}$;
6:	if $ Obj^{(t)} - Obj^{(t-1)} \leq \varepsilon$, Return $\mathbf{s}^* = \mathbf{s}^{(t)}$, $\mathbf{w}^* = \mathbf{w}_n^{(t-1)}$;
7:	Otherwise, return to Step 2;

IV. SIMULATION ANALYSIS

In this section, we consider a distributed system with three nodes, assume the designed transmit waveform is used for node 1. The signal bandwidth is $B = 50\text{MHz}$, the total number of subpulses is $M = 200$. The number of communication subpulses is $N_p = 2, 50, 100$. Specifically, the number of communication subpulses transmitted to node 2 and node 3 are $N_{p_1} = 1, 25, 50$ and $N_{p_2} = 1, 25, 50$, respectively. During information modulation, the phase-difference sets for node 2 and node 3 are $\mathcal{U}_2: \{0, \pi\}$ and $\mathcal{U}_3: \{\pi/2, 3\pi/2\}$, respectively. The phase protection interval is $\kappa = 0.3$.

Fig.3 (a) illustrates that, it can be observed that the objective function value monotonically decreases with the increase of iterations and converges to a fixed value, verifying the effectiveness and convergence of the proposed IBCD-DA algorithm. Under a fixed total number of subpulses, a smaller number of communication subpulses results in a lower converged objective function value, indicating that fewer communication subpulses provide greater DoFs for waveform design. Fig.3 (b) illustrates the phase distribution of the signal. The subpulse phases within the black and blue dashed ellipses are used to transmit communication information to nodes 2 and 3, respectively, achieving multi-user communication functionality. Fig.3 (c) presents the Symbol Error Rate (SER) analysis for $N_p = 2$ and $N_p = 100$, It can be observed that as the number of communication subpulses decreases, the SER performance

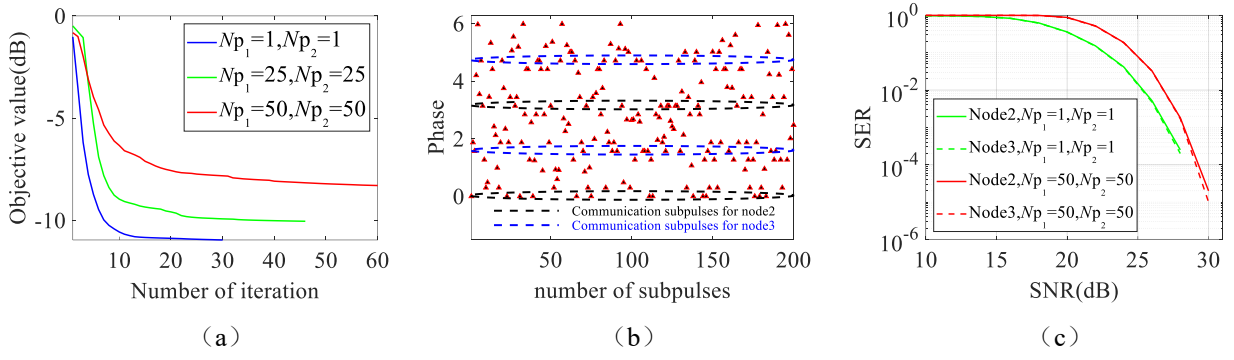


Fig.3. Optimization results and SER analysis . (a) Objective function value versus iteration number; (b) Phase distribution versus subpulse number when $N_{p_1} = 25, N_{p_2} = 25$; (c) SER performance analysis.

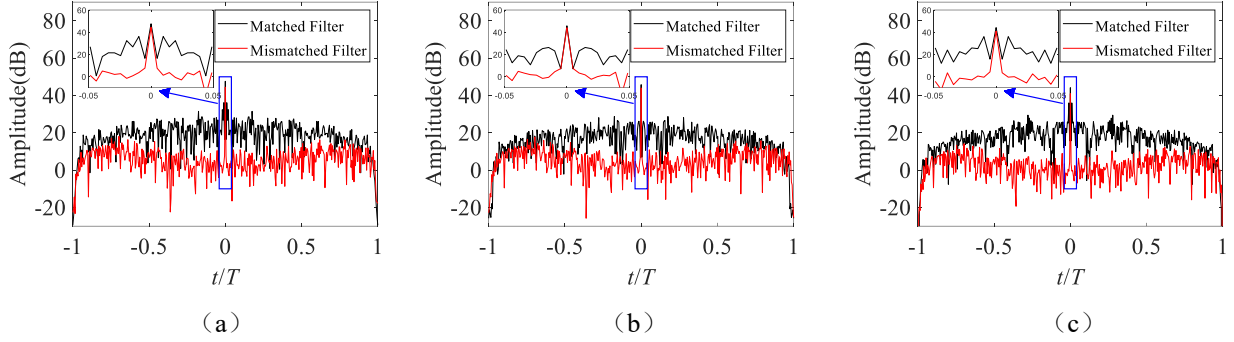


Fig.4. Mismatched filter results. (a) node 1; (b) node 2; (c) node 3.

improves. This is because, when each communication subpulse is detected with an identical detection probability, increasing the number of communication subpulses leads to a higher SER. Furthermore, it can also be observed that at high Signal-to-Noise Ratios (SNR), information can be effectively transmitted to two nodes with a very low SER. This paper investigates a radar-centric distributed DFRC architecture, in which the high transmit power and one-way propagation attenuation ensure operation in a high-SNR regime, thereby supporting multi-user communication.

Fig. 4 shows the matched and mismatched filtering results at one node. By jointly optimizing the transmitted waveform and the mismatched filters, the cross-correlation sidelobe level is significantly reduced, thereby improving detection accuracy for adjacent strong and weak targets. This gain is achieved at the cost of SNR loss, which is approximately 3 dB on average.

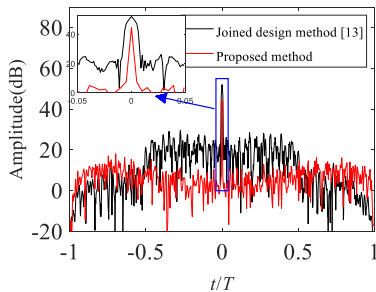


Fig.5. Comparison results

To the best of our knowledge, there is limited research on distributed wideband integrated signal design, a wideband radar signal design method in [13] is considered as a

benchmark. Fig. 5 illustrates that, compared to the waveform designed in [13], the waveform proposed in this paper reduces the sidelobe level, significantly improving the ability to detect weak targets in strong clutter or dense target environments. In contrast, the waveform designed in [13] increases the SNR by extending the cross-correlation mainlobe, which, however, leads to a reduction in range resolution.

V. CONCLUSION

This paper proposes a joint integrated wideband waveform and mismatched filter design method for distributed DFRC systems. By minimizing the ratio of integrated cross-correlation sidelobe levels to mainlobe levels across multiple receiving nodes under the PIP modulation constraints, a non-convex optimization problem with multiple constraints is formulated. To solve this problem efficiently, an IBCD-DA algorithm is developed, which guarantees a monotonically decreasing objective function and convergence within polynomial time. Compared with existing wideband radar waveform design methods, the proposed scheme achieves lower sidelobe levels, maintains high range resolution, and enables efficient multi-user communication, as validated by simulation results.

ACKNOWLEDGMENT

This work was supported in part by the National Natural Science Foundation of China (62571099), in part by the Natural Science Foundation of Sichuan Province under Grant 2026NSFSC1418 and in part by the Science and Technology Research Program of Chongqing Municipal Education Commission (KJQN202501240).

REFERENCES

- [1] F. Liu, A. Petropulu, H. Griffiths, L. Hanzo, and C. Masouros, "Joint radar and communication design: Applications, state-of-the-art, and the road ahead," *IEEE Trans. Commun.*, vol. 68, no. 6, pp. 3834–3862, Jun. 2020.
- [2] A. Hassanien, B. Himed, E. Aboutanios, and M. Amin, "Dual-function radar communication systems: A solution to the spectrum congestion problem," *IEEE Signal Process. Mag.*, vol. 36, no. 5, pp. 115–126, Sep. 2019.
- [3] Y. Liu, G. Liao, H. Li, M. Jiang, and Y. Chen, "Distributed integrated waveform design and information acquisition in electromagnetic space," *Bull. Natl. Nat. Sci. Found. China*, vol. 35, no. 5, pp. 701–707, 2021.
- [4] Y. Guo, X. Huang, K. Wu, J. Zhang, and J. Yuan, "Reliable frequency-hopping MIMO radar-based communications with multi-antenna receiver," *IEEE Trans. Commun.*, vol. 69, no. 8, pp. 5502–5513, Aug. 2021.
- [5] C. Tsinos, A. Arora, S. Chatzinotas, and B. Ottersten, "Joint transmit waveform and receive filter design for dual-function radar-communication systems," *IEEE J. Sel. Topics Signal Process.*, vol. 15, no. 6, pp. 1378–1392, Nov. 2021.
- [6] A. Swindlehurst, M. Li, Q. Liu, and R. Liu, "Joint waveform and filter designs for STAP-SLP-based MIMO-DFRC systems," *IEEE J. Sel. Areas Commun.*, vol. 40, no. 6, pp. 1918–1931, Jun. 2022.
- [7] Z. Liu, "Waveform research on integration of radar and communication," Ph.D. dissertation, Beijing Univ. Technol., Beijing, China, 2015.
- [8] S. Blunt, J. Jakabosky, J. Metcalf, C. Sahin, and P. McCormick, "A novel approach for embedding communication symbols into physical radar waveforms," in *Proc. IEEE Radar Conf.*, Seattle, WA, USA, May 2017, pp. 1498–1503.
- [9] G. Cui, L. Kong, J. Yang, X. Yu, and S. Lu, "Dual-use unimodular sequence design via frequency nulling modulation," *IEEE Access*, vol. 6, pp. 62470–62481, 2018.
- [10] J. Yang, X. Yu, G. Cui, and L. Kong, "Dual-use signal design for radar and communication via ambiguity function sidelobe control," *IEEE Trans. Veh. Technol.*, vol. 69, no. 9, pp. 9781–9794, Sep. 2020.
- [11] Y. Liu, X. Yao, X. Yu, J. Zhang, and G. Cui, "Dual function radar and communication waveform design based on sub-pulse hybrid modulation," *Journal of Radars*, vol. 14, no. 4, pp. 896–914, Jul. 2025.
- [12] X. Yao, Y. Liu, H. Qiu, G. Cui, X. Yu, and Z. Zhang, "Dual-use baseband signal design for radCom with position index and phase modulation," *Signal Process.*, vol. 209, no. 109015, 2023.
- [13] Y. Wu, T. Fan, X. Yu, R. Tan, G. Cui, and P. Zhu, "An extended target signal integration method via mainlobe broadening," *Signal Process.*, vol. 216, pp. 1–5, 109298, 2023.
- [14] T. Wei, H. Meng, X. Wang, and Y. Liu, "Radar phase-modulated waveform design for extended target detection," *Tsinghua Science and Technology*, vol. 16, no. 4, pp. 364–370, Apr. 2011.
- [15] X. Yu, X. Yao, J. Yang, L. Zhang, L. Kang, and G. Cui, "Integrated waveform design for MIMO radar and communication via spatio-spectral modulation," *IEEE Trans. Signal Process.*, vol. 70, pp. 2293–2305, Apr. 2022.
- [16] J. Yang, A. Aubry, A. De Maio, X. Yu, and G. Cui, "Multispectrally constrained transceiver design against signal-dependent interference," *IEEE Trans. Signal Process.*, vol. 70, pp. 1320–1332, 2022.
- [17] J. P. Crouzeix, J. A. Ferland, and S. Schaible, "An algorithm for generalized fractional programs," *J. Optim. Theory Appl.*, vol. 47, no. 1, pp. 35–49, Sep. 1985.